

ACRME Placement Scoring Weights — Complete Reference

Project: Azure Capacity Reservation Management Engine (ACRME)

Classification: Technical Reference

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Overview

The ACRME placement engine uses **three environment-type-specific scoring formulas** (PS_Prod, PS_NonProd, PS_DR) to rank Azure regions for Prod, NonProd/CVAL, and DR environment placement. All three formulas share the same **default weight values**:

- = 0.30 (CRG Capacity)
- = 0.20 (Quota Headroom)
- = 0.25 (Distribution Fairness)
- = 0.15 (DR/Overflow Health)
- = 0.10 (Zone Diversity)

However, and have **different semantic meanings** per environment type — they measure different signals depending on whether you’re scoring a Prod, NonProd, or DR region.

Weight Definitions by Component

(Alpha) — CRG Capacity Signal [Weight: 0.30]

Purpose: Measures available capacity in the region’s CRG most relevant to the environment being placed.

Semantic per environment:

Environment	Measures	Formula
PS_Prod	NonProd CRG overflow headroom relative to Prod CRG size. High = the associated NonProd CRG has capacity to absorb overflow if Prod grows.	$\frac{\text{nonprod_crg_effective_free}}{\text{prod_crg_quantity}}$

Environment	Measures	Formula
PS_NonProd	Fraction of NonProd CRG that is truly available (free_slots minus dr_overflow_reserve). This is the primary capacity signal for NonProd placement.	$\frac{\text{nonprod_crg_effective_free}}{\text{nonprod_crg_quantity}}$
PS_DR	Fraction of DR CRG capacity that is uncommitted. High = more room for additional DR assignments.	$\frac{\text{dr_crg_free_slots}}{\text{dr_crg_quantity}}$

Why is the highest weight (0.30): Capacity is the most critical constraint. A region with zero headroom cannot accept new placements regardless of quota or distribution. is highest when capacity is tight; can be reduced to 0.20 once regions reach steady-state utilization (40–60%).

(Beta) — Quota Headroom [Weight: 0.20]

Purpose: Measures available Azure subscription/group quota headroom for the relevant CRG.

Semantic per environment:

Environment	Measures	Formula
PS_Prod	Prod group quota headroom ratio	$\frac{\text{prod_crg_quota_headroom_vcpu}}{\text{prod_crg_quota_limit_vcpu}}$
PS_NonProd	NonProd floor-adjusted quota headroom (effective ceiling minus used)	$\frac{\text{nonprod_crg_quota_headroom_vcpu}}{\text{nonprod_crg_quota_limit_vcpu}}$
PS_DR	DR quota headroom within the DR floor allocation	$\frac{\text{dr_crg_quota_headroom_vcpu}}{\text{dr_crg_quota_limit_vcpu}}$

Why = 0.20: Quota exhaustion silently blocks CR creation and VM deployment. ensures the engine directs placements toward regions with quota

headroom. Lower than `beta` because quota can be requested (takes time but is not a hard ceiling like physical capacity).

Note: All `beta` calculations read from **quota-group-level fields** (two-group model: Prod-only group, shared NonProd+DR group), not legacy per-subscription quota. See ADR-002.

(Gamma) — Distribution Fairness [Weight: 0.25]

Purpose: Acts as a **self-correcting load balancer**. Rewards regions with fewer customers already placed, driving even distribution across regions over time.

Semantic (same for all environments):

$$\gamma = 1 - (\text{customer_count_in_this_region} / \text{total_customers})$$

Environment	Measures	Formula
PS_Prod	Prod region fairness	$1 - (\text{consumer_count_prod} / \text{total_customers})$
PS_NonProd	NonProd region fairness	$1 - (\text{consumer_count_nonprod} / \text{total_customers})$
PS_DR	DR region fairness	$1 - (\text{consumer_count_dr} / \text{total_customers})$

Why $\gamma = 0.25$: Equal to capacity in importance. As a region accumulates customers its `beta` score drops, redirecting future placements away from it until all regions reach roughly equal count (steady-state expected count per region per env type = `total_customers / 3`). Increase `beta` toward 0.35 if score convergence results in uneven loading; decrease toward 0.15 if capacity constraints should dominate fairness.

Note: The PRR (Section 28) updates Distribution to use **demand units** rather than customer count: $1 - (\text{Region_Assigned_Demand} / \text{Total_Assigned_Demand})$.

(Delta) — DR/Overflow Health [Weight: 0.15]

Purpose: Environment-specific **resilience signal** — measures DR coverage readiness (Prod/DR) or overflow capacity health (NonProd).

Semantic per environment:

Environment	Measures	Formula	Intent
PS_Prod	DR CRG coverage readiness of this region’s own DR CRG. When scoring a Prod region, captures the health of the region’s DR CRG — a Prod region whose DR CRG is well-covered scores higher.	$\text{dr_crg_coverage_ratio}$	Prod placement favors regions with healthy DR capacity
PS_NonProd	Overflow capacity health — intentionally shares the same ratio as . Rewards regions genuinely under-utilised from both a NonProd-placement and a DR-overflow perspective.	$\frac{\text{nonprod_crg_effectiveness}}{\text{nonprod_crg_quantity}}$ (same as)	NonProd placement quantifies regions with DR overflow headroom

Environment	Measures	Formula	Intent
PS_DR	Coverage ratio health relative to target. Scores 1.0 when coverage $\geq \text{dr_ratio_max}$ (0.40). Proportionally below 1.0 for coverage between dr_ratio_min (0.30) and dr_ratio_max (e.g. $0.35/0.40 = 0.875$).	$\min(1.0, \text{dr_crg_coverage_patient} / \text{dr_ratio_max})$	DR patient drives buffer margins toward target

Why is different per env: Each environment has a distinct resilience need. Prod cares about its paired DR region’s readiness; NonProd cares about overflow headroom it can yield to DR during a crisis; DR cares about its own coverage health.

Why = 0.15: Lower than // because it’s a forward-looking health signal rather than an immediate placement blocker. Increase toward 0.20 if DR buffer violations appear; decrease if capacity/distribution should dominate.

Trade-off (PS_NonProd): duplicates intentionally — this was flagged in the PRR (Section 28) as double-counting. The corrected pilot formula proposes revised NonProd weights: $\text{PS_NonProd} = 0.35 \cdot \text{Capacity} + 0.25 \cdot \text{Quota} + 0.25 \cdot \text{Distribution} + 0.05 \cdot \text{DR_Overflow_Integrity} + 0.10 \cdot \text{Zones}$ to reduce the duplication.

(Epsilon) — Zone Diversity [Weight: 0.10]

Purpose: Rewards regions with more availability zones, ensuring placed workloads can leverage Azure’s zone-redundancy features.

Semantic (same for all environments):

$$= \text{az_count} / 3$$

Normalizes to 1.0 for a 3-zone region (maximum diversity); proportionally lower for 2-zone (0.667) or 1-zone (0.333) regions.

Environment	Measures	Formula
All (PS_Prod, PS_NonProd, PS_DR)	Zone diversity	az_count / 3

Why = 0.10 (lowest weight): Zone diversity is desirable but not a placement blocker. Hard constraint **HC-5 ZONE_AVAILABILITY** already enforces a minimum zone count — acts as a tiebreaker that favors 3-zone regions when all else is equal. Increase toward 0.15 if zone diversity is a critical operational requirement; keep at 0.10 if capacity/distribution dominate.

Complete Formula Definitions

PS_Prod(r) — Production Region Score

```

PS_Prod(r) =
    0.30 × (nonprod_crg_effective_free / prod_crg_quantity)           ← : NonProd overflow
  + 0.20 × (prod_crg_quota_headroom_vcpu / prod_crg_quota_limit_vcpu) ← : Prod quota headr
  + 0.25 × (1 - consumer_count_prod / total_customers)               ← : Prod distributio
  + 0.15 × dr_crg_coverage_ratio                                       ← : DR readiness of
  + 0.10 × (az_count / 3)                                              ← : zone diversity

```

Guards: - If prod_crg_quantity == 0 → PS_Prod = 0.0 - If prod_crg_quota_limit_vcpu == 0 → term = 0.0 - If dr_crg_coverage_ratio is None → term = 0.0 (no DR CRG established yet)

PS_NonProd(r) — Non-Production / CVAL Region Score

Current formula (with intentional / duplication):

```

PS_NonProd(r) =
    0.30 × (nonprod_crg_effective_free / nonprod_crg_quantity)       ← : effective M
  + 0.20 × (nonprod_crg_quota_headroom_vcpu / nonprod_crg_quota_limit_vcpu) ← : NonProd quo
  + 0.25 × (1 - consumer_count_nonprod / total_customers)           ← : NonProd dis
  + 0.15 × (nonprod_crg_effective_free / nonprod_crg_quantity)       ← : overflow ca
  + 0.10 × (az_count / 3)                                              ← : zone divers

```

Proposed pilot formula (corrected to avoid double-counting):

```

PS_NonProd(r) =
    0.35 × (nonprod_crg_effective_free / nonprod_crg_quantity)       ← : capacity (i
  + 0.25 × (nonprod_crg_quota_headroom_vcpu / nonprod_crg_quota_limit_vcpu) ← : quota headr
  + 0.25 × (1 - consumer_count_nonprod / total_customers)           ← : distributio
  + 0.05 × (nonprod_crg_effective_free / nonprod_crg_quantity)       ← : DR overflow
  + 0.10 × (az_count / 3)                                              ← : zone divers

```

Guards: - If `nonprod_crg_quantity == 0` $\rightarrow$ `PS_NonProd = 0.0` - If `nonprod_crg_quota_limit_vcpu == 0` $\rightarrow$ `term = 0.0`

PS_DR(r) — Disaster Recovery Region Score

`PS_DR(r) =`
`0.30 × (dr_crg_free_slots / dr_crg_quantity)` ← : DR CRG head
`+ 0.20 × (dr_crg_quota_headroom_vcpu / dr_crg_quota_limit_vcpu)` ← : DR quota head
`+ 0.25 × (1 - consumer_count_dr / total_customers)` ← : DR distribut
`+ 0.15 × min(1.0, dr_crg_coverage_ratio / dr_ratio_max)` ← : coverage rat
`+ 0.10 × (az_count / 3)` ← : zone diversi

Guards: - If `dr_crg_quantity == 0` $\rightarrow$ `PS_DR = 0.0` (no DR CRG — region ineligible) - If `dr_crg_quota_limit_vcpu == 0` $\rightarrow$ `term = 0.0` - If `dr_crg_coverage_ratio` is `None` $\rightarrow$ `term = 0.0`

Weight Summary Table

Component	Symbol	Weight	PS_Prod Measures	PS_NonProd Measures	PS_DR Measures
CRG Capacity		0.30	NonProd overflow headroom	NonProd effective headroom	DR CRG free capacity
Quota Headroom		0.20	Prod group quota ratio	NonProd floor-adjusted quota	DR quota within floor
Distribution		0.25	Prod region fairness	NonProd region fairness	DR region fairness
DR/Overflow Health		0.15	DR coverage readiness	Overflow capacity (same as)	Coverage vs target ratio
Zone Diversity		0.10	<code>az_count / 3</code>	<code>az_count / 3</code>	<code>az_count / 3</code>

Clamping & Score Range

Every raw component ratio is **clamped** before weighting:

`Clamp(x) = max(0, min(1, x))`

This ensures each component contributes a value in the range $[0.0, 1.0]$, and the final weighted score for any region is:

$PS(r) \in [0.0, 1.0]$

A uniform jitter of ± 0.01 is added as a tiebreaker.

Tuning Guidance

When to adjust (Capacity)

- **Increase toward 0.35–0.40** when capacity is tight and regions are unevenly provisioned.
- **Reduce toward 0.20** once regions reach steady-state utilization (40–60%) and distribution becomes the primary concern.

When to adjust (Distribution)

- **Increase toward 0.35** if score convergence results in uneven region loading (some regions accumulate too many customers).
- **Decrease toward 0.15** if capacity constraints should dominate fairness (tight capacity environments).

When to adjust (DR/Overflow Health)

- **Increase toward 0.20** if DR buffer violations or coverage gaps appear frequently.
- **Reduce toward 0.10** if capacity/distribution should dominate and DR health is consistently good.

When to adjust (Zone Diversity)

- **Increase toward 0.15** if zone diversity is a critical operational requirement (e.g., all placements must strongly favor 3-zone regions).
- **Keep at 0.10** (default) if capacity, quota, and distribution are the dominant concerns.

Default recommendation: Start with the default weights ($w_{cap}=0.30$, $w_{dist}=0.20$, $w_{dr}=0.25$, $w_{div}=0.15$, $w_{zone}=0.10$) for the pilot. After 30–50 customer placements, analyze the score breakdown data in `OperationRecord` and adjust incrementally based on observed convergence behavior.

Source References

- **Multi-Region Placement Design** (`docs/research/multi_region_placement_design.md`)
— lines 449–531, full formula definitions + weight table

- **Design Change Summary** (docs/research/design_change_summary.md)
— CR/CRG-7, lines 153–193, formula rationale
 - **Production Readiness Review** (docs/acrme_production_readiness_review_and_architecture.m
— Section 28 lines 1095–1160, corrected scoring model
 - **ADR-001 Region Selection** (docs/adr/acrme_adr_001_region_selection.md)
— corrected scoring model section
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Evidence Tags

- [Documented] — Defined in Multi-Region Placement Design and PRR
Section 28
 - [Decided] — Decision Log D9 (env-type-specific formulas)
 - [Assumed] — Default weights retained for pilot comparison; revised
weights proposed but not empirically validated
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Document Status: Reference

Next Review: After pilot Phase 4 (policy tuning with 30–50 placements measured)